

Onkaar Grewal

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EDUCATION

McMaster University

Engineering I | Focus: Engineering Physics (Co-op)

Hamilton, ON

Sep 2025 – Present

EXPERIENCE

Engineering Design & Drafting Intern

Atlantic Industries Limited

Cambridge, ON

January 2023 – June 2023

- Produced high-accuracy 2D AutoCAD drawings for structural civil engineering projects, transitioning from administrative support to technical drafting.
- Collaborated with university co-op students on large-scale infrastructure designs and assisted in document management and drawing revisions.

Administrative Project Support

YMCA of Three Rivers

Ontario, Canada

March 2025 – Present

- Coordinate board-level initiatives and events across 6 municipal locations, managing project documentation and scheduling for regional leadership.
- Facilitate cross-location communication for the Board of Directors to ensure efficient execution of community-focused projects.

First Robotics Competition Mentor

St. Benedict Catholic Secondary School

Cambridge, ON

September 2023 – January 2025

- Mentored mechanical sub-teams in CAD prototyping and fabrication of competitive 120lb robots, guiding drivetrain and intake design layout.
- Instructed junior students on shop safety and the safe operation of power tools and industrial machinery.

PROJECTS

Custom High-Temp Enclosed 3D Printer | *SolidWorks, Klipper, Fourier's Law*

2024 – Present

- Designing a custom CoreXY printer using composite wall thermal analysis to maintain a 100°C+ chamber for high-performance polymers.
- Integrating Rapido 2 UHF and Manta M8P components with Klipper firmware for advanced resonance compensation and motion tuning.

Custom Weathering Station Hub | *Sensors, Data Logging, Mechanical Design*

2024

- Developed an autonomous weather hub utilizing a sensor suite for real-time monitoring of humidity, temperature, wind rate, and rain gauge levels.
- Engineered a durable mechanical housing to protect sensitive electronics from environmental exposure while ensuring accurate sensor readings.

Wheelchair-Mounted Assistive Launcher | *SolidWorks, Granta EduPack, Python*

2025

- Led mechanical CAD design of a launcher achieving a 6m+ range, utilizing Granta EduPack for structural material selection to benefit a client with arthritis.
- Coordinated technical design integration across a multidisciplinary team within McMaster's Engineering Design course (ENGINEER 1P13).

High-Velocity Payload Transport System | *AutoCAD, Mechanical Design, Hydraulics*

2023

- Designed a hydraulic scale system for high-speed payload transport, achieving the fastest completion time in the cohort.
- Engineered a capture claw using surface-area optimization to facilitate rapid deceleration of fragile payloads.

TECHNICAL SKILLS

Design & CAD: SolidWorks, Autodesk Inventor, AutoCAD, Fusion 360, KiCad, Blender

Programming & Tools: Python, G-Code, C/C++, MATLAB, LaTeX, Klipper, Granta EduPack, OrcaSlicer

Hardware & Fabrication: 3D Printing, Soldering, Stepper Motor Systems, Hydraulics, Raspberry Pi, Power Tools